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Zero-state response of a continuous LTI system

- The system responds with zero state
- Calculation of convolutional integrals
- The nature of convolutional integrals

1. System zero-state response

If the input signal is x(t), the impulse response of the continuous-time LTI system is h(t), then the zero-state response of the system $y_{zs}(t)$ is:

$$y_{zs}(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau \qquad \text{(convolution)}$$

It can be seen that the zero-state response of a continuous LTI system is a convolutional integral of the input signal and the impulse response of the system, which reveals the interaction mechanism between the signal and the system in the time domain.

2. Calculation of convolutional points

Analysis method: It is calculated directly from the expression of the convolutional integral

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

If the signals x(t) and h(t) can be expressed analytically functionally. The analytical method can be used to calculate convolutional integrals.

[Example] calculate $x(t)*h(t), x(t)=\mathrm{e}^{-t}u(t), h(t)=u(t)$

Solution: Defined by convolution

$$x(t)*h(t)=\int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

$$=\int_{-\infty}^{\infty} \mathrm{e}^{-\tau}u(\tau)u(t-\tau)\mathrm{d}\tau$$

$u(\tau)u(t-\tau)$ to get

$$=\begin{cases}\int_0^t \mathrm{e}^{-\tau}\mathrm{d}\tau=1-\mathrm{e}^{-t}, & t>0\\ 0 & t<0\end{cases}$$

Then, get

$$x(t)*h(t)=(1-\mathrm{e}^{-t})u(t)$$

[Example] Find convolutional points of $2e^{-2t}u(t) * 3e^{-t}u(t)$

Solution: $2e^{-2t}u(t) * 3e^{-t}u(t)$

$$= \int_{-\infty}^{+\infty} 2e^{-2\tau}u(\tau) \cdot 3e^{-(t-\tau)}u(t-\tau)\mathrm{d}\tau$$

$$= \begin{cases} 6\int_0^t e^{-2\tau}e^{-(t-\tau)}\mathrm{d}\tau & t > 0 \\ 0 & t < 0 \end{cases}$$

$$= \begin{cases} 6e^{-t}\int_0^t e^{-\tau}\mathrm{d}\tau & t > 0 \\ 0 & t < 0 \end{cases} = 6(e^{-t} - e^{-2t})u(t)$$

$$e^{\alpha t}u(t) * e^{\beta t}u(t) = \begin{cases} \dfrac{1}{\beta - \alpha}(e^{\beta t} - e^{\alpha t})u(t) & \alpha \neq \beta \\ te^{\alpha t}u(t) & \alpha = \beta \end{cases}$$

2. Calculation of convolutional points

Graphical method: $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$

Steps to calculate convolutional integrals graphically:

(1) Change the independent variables in $x(t)$ and h(t) from t to τ

(2) Flip one of the signals to $h(-\tau)$ and then pan t to get $h(t-\tau)$

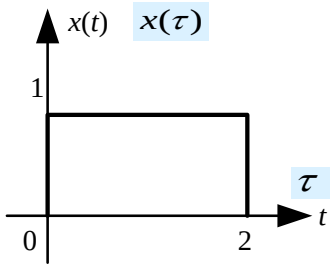
$$h(\tau) \xrightarrow{\text{翻转}} h(-\tau) \xrightarrow{\text{平移}t} h(-(\tau - t)) = h(t - \tau)$$

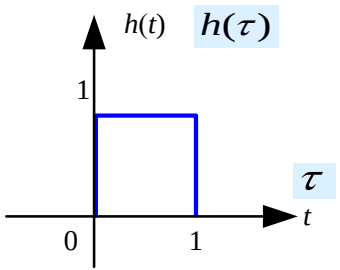
(3) Multiply $x(t)$ by $h(t-\tau)$ to integrate the signal after the product.

(4) Constantly changing the amount of translation t, calculating the integral of $x(t)$ $h(t-\tau)$ respectively.

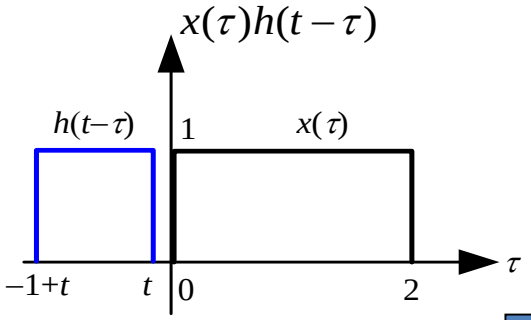
[Example] calculation $y(t) = x(t) * h(t)$

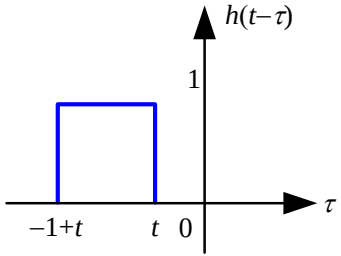
Solution:





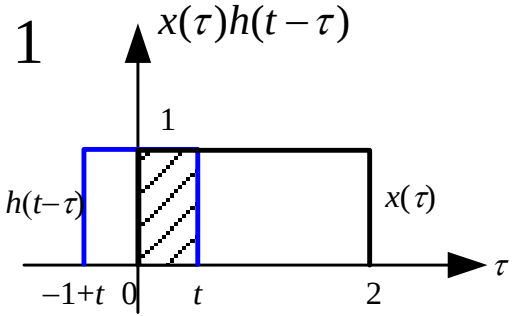
1) $t < 0$



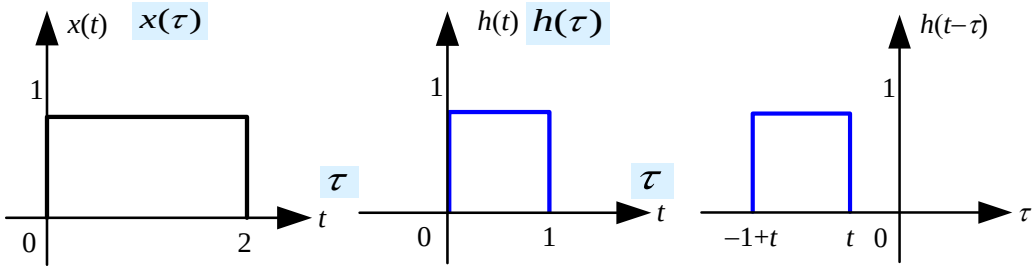


$$y(t) = 0$$

2) $0 \le t < 1$

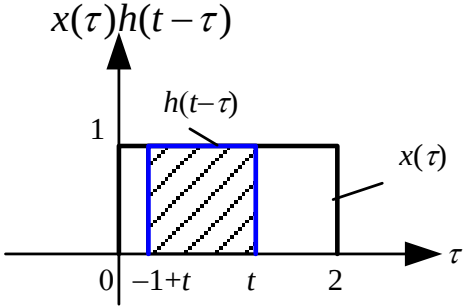


$$y(t) = \int_0^t 1 \times 1 \, \mathrm{d}\tau = t$$



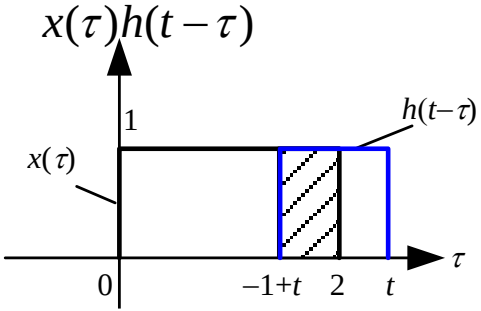
Solution:

c) $1 \le t < 2$



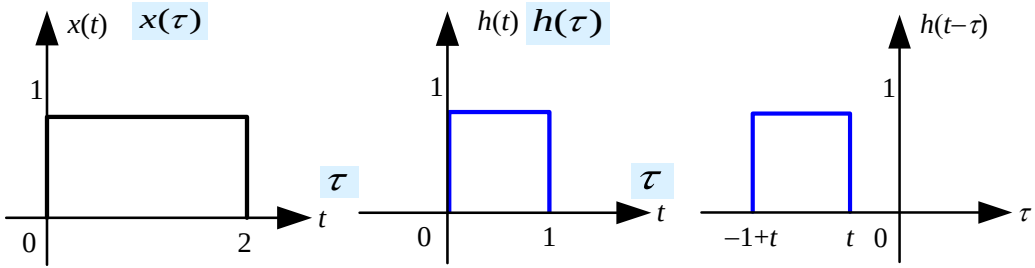
$$\Rightarrow y(t) = \int_{-1+t}^{t} 1 \times 1 \, \mathrm{d}\tau = 1$$

d) $2 \le t < 3$



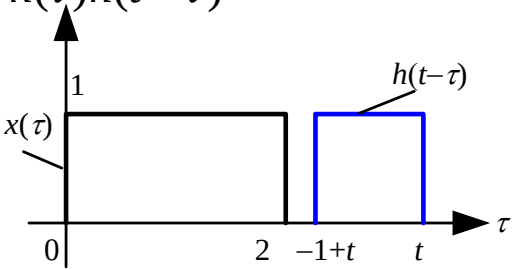
$$\Rightarrow y(t) = \int_{-1+t}^{2} 1 \times 1 \, \mathrm{d}\tau = 3 - t$$

Solution:



e) $3 \le t$

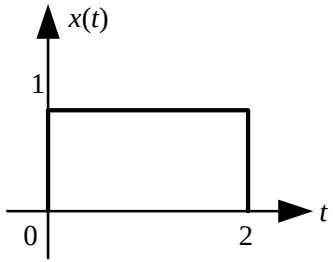
$x(\tau)h(t-\tau)$

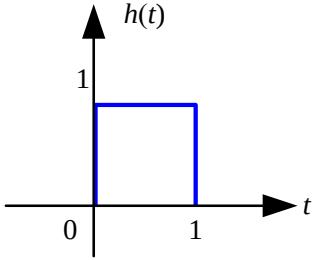


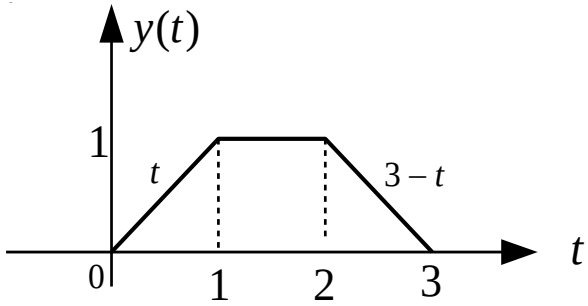
$$y(t) = 0$$

[Example] calculation $y(t) = x(t) * h(t)$

Solution: In summary, it can be obtained:







The convolution of two unequal-width rectangular pulses is an isosceles trapezoidal signal, the lower floor width is the sum of the two signal widths, the upper bottom width is the difference between the two signal widths, and the height is the maximum coincidence area of the two signals.

The convolution of two signals, the convolution result is still one signal. The starting point of the signal is equal to the sum of the beginnings of the two signals, and the end point is equal to the sum of the end points of the two signals.

3. The nature of convolutional integrals

(1) Commutative law: $x_1(t) * x_2(t) = x_2(t) * x_1(t)$

(2) Distributive law: $[\, x_1(t) + x_2(t) \,] * x_3(t) = x_1(t) * x_3(t) + x_2(t) * x_3(t)$

(3) Binding law : $[\, x_1(t) * x_2(t) \,] * x_3(t) = x_1(t) * [\, x_2(t) * x_3(t) \,]$

3. The nature of convolutional integrals

※ Translation characteristics:

If $x_1(t) * x_2(t) = y(t)$, then $x_1(t-t_1) * x_2(t-t_2) = y(t-t_1-t_2)$

$$x_1(t-2) * x_2(t-2) = y(t-4)$$

$$x(t) * \delta(t) = x(t)$$

$$x(t) * \delta(t - t_0) = x(t - t_0)$$

$$x(t - t_1) * \delta(t - t_2) = x(t - t_1 - t_2)$$

3. The nature of convolutional integrals

※ **Equivalent characteristics:** $x_1(t) * x_2(t) = x_1^{(-1)}(t) * x_2'(t) = x_1'(t) * x_2^{(-1)}(t)$

※ **Differential properties:**

If $x(t) * h(t) = y(t)$, then $x'(t) * h(t) = x(t) * h'(t) = y'(t)$

$$x(t) * \delta'(t) = x'(t)$$

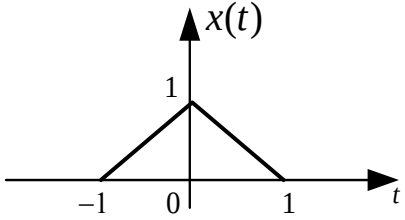
※ **Integral characteristics:**

If $x(t) * h(t) = y(t)$, then $x^{(-1)}(t) * h(t) = x(t) * h^{(-1)}(t) = y^{(-1)}(t)$

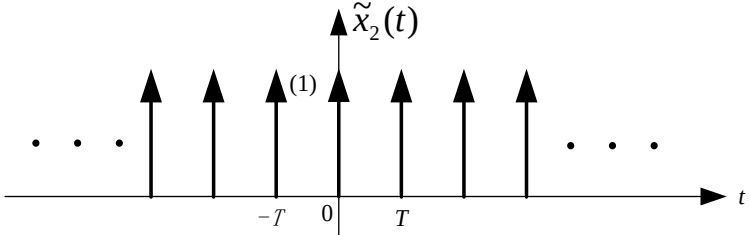
$$x(t) * u(t) = \int_{\infty}^{t} x(\tau)\mathrm{d}\tau = x^{(-1)}(t)$$

3. The nature of convolutional integrals

Calculate the convolution of the trigonometric wave $x_1(t)$ and the periodic impulse string signal $\tilde{x}_2(t)$ when T=2 and T=1 are calculated.



Triangular waves



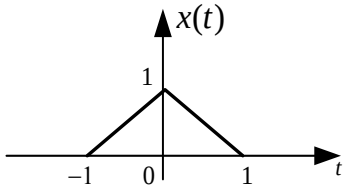
Periodic impulse string signal

Periodic impulse string signal $\tilde{x}_2(t) = \cdots + \delta(t+T) + \delta(t) + \delta(t-T) + \cdots$

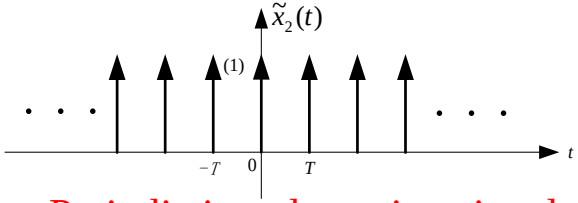
Take advantage of the translation characteristics of convolution $x(t) * \delta(t-T) = x(t-T)$

To find the convolution of the triangular wave and each impulse signal in $\tilde{x}_2(t)$, you can use the convolutional distribution law to add up the convolutional results.

Calculate the convolution of the trigonometric wave $x_1(t)$ and the periodic impulse string signal $\tilde{x}_2(t)$ when T=2 and T=1 are calculated.



Triangular waves

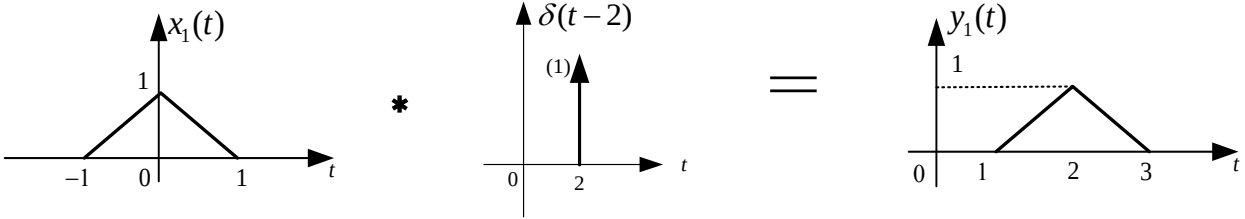


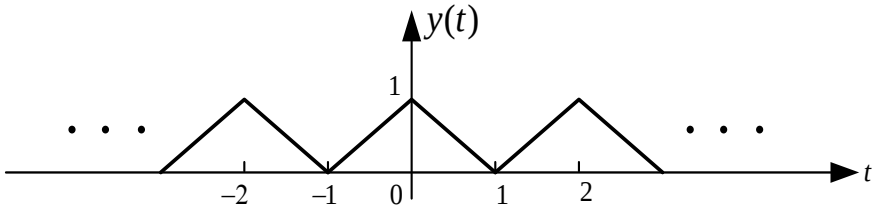
Periodic impulse string signal

T=2

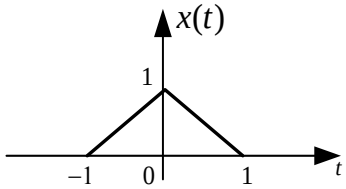
At T=2, the periodic impulse string signal $\tilde{x}_2(t) = \cdots + \delta(t+2) + \delta(t) + \delta(t-2) + \cdots$

By $x(t) * \delta(t - T) = x(t - T)$

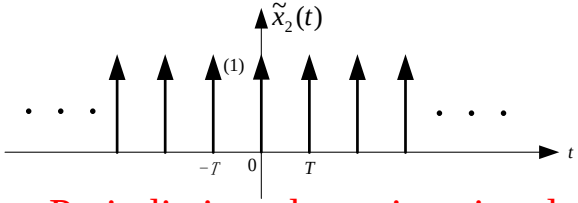




Calculate the convolution of the trigonometric wave $x_1(t)$ and the periodic impulse string signal $\tilde{x}_2(t)$ when T=2 and T=1 are calculated.



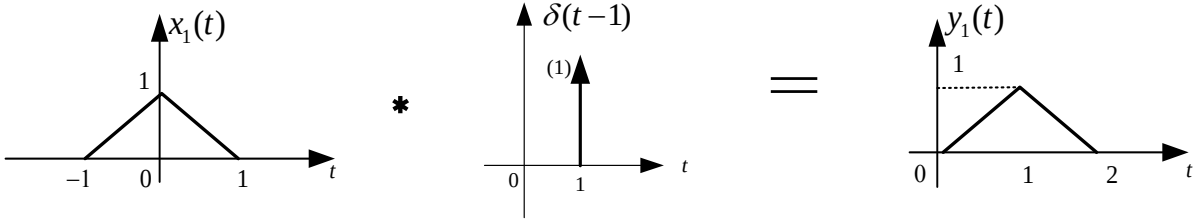
Triangular waves

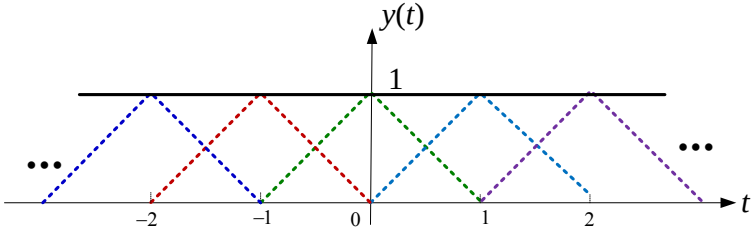


Periodic impulse string signal

T=1

At T=1, the periodic impulse string signal $\tilde{x}_2(t) = \cdots + \delta(t+1) + \delta(t) + \delta(t-1) + \cdots$

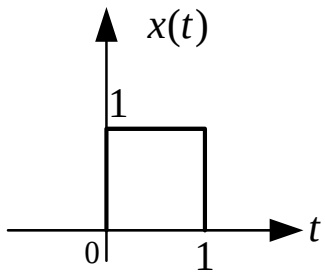


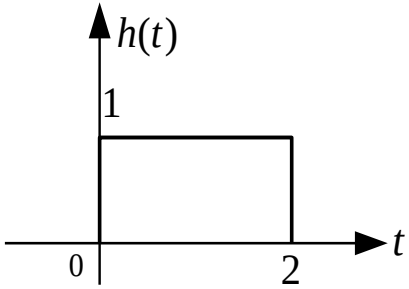


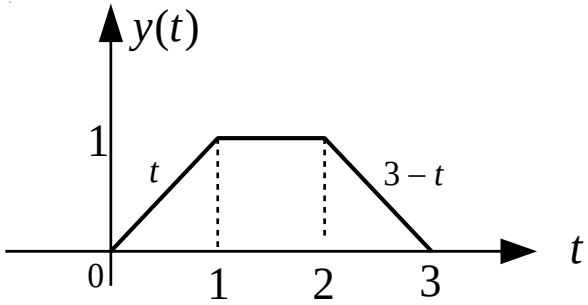
3. The nature of convolutional integrals

[Example] Using the translation feature and $u(t) * u(t) = r(t)$,

calculate $y(t) = x(t) * h(t)$







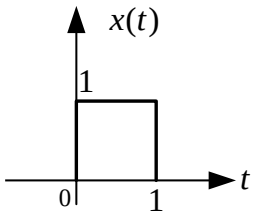
Solution: $y(t) = x(t) * h(t) = [\, u(t) - u(t-1) \,] * [\, u(t) - u(t-2) \,]$

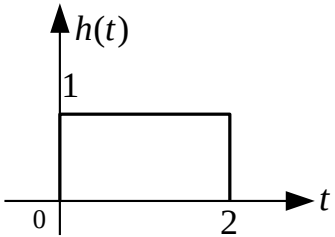
$$= u(t)*u(t) - u(t-1)*u(t) - u(t)*u(t-2) + u(t-1)*u(t-2)$$

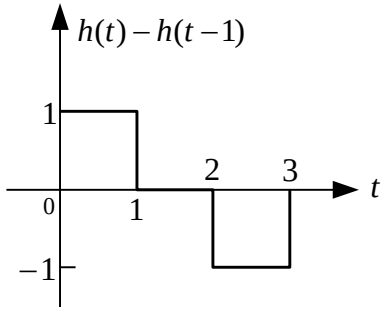
$$= r(t) - r(t-1) - r(t-2) + r(t-3)$$

3. The nature of convolutional integrals

[Example] Using equivalent characteristics, calculations $y(t) = x(t) * h(t)$



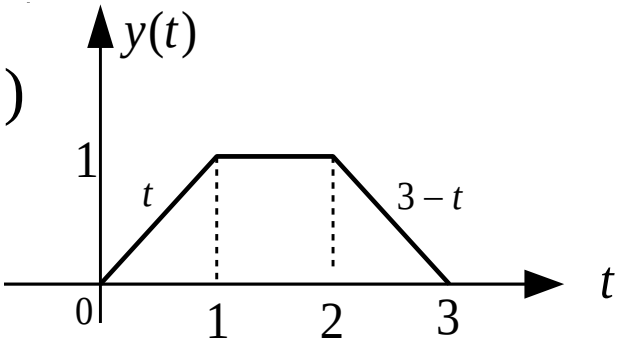




Solution: $x'(t) = \delta(t) - \delta(t-1)$

$$y'(t) = x'(t) * h(t) = h(t) - h(t-1)$$

$$y(t) = \int_{-\infty}^{t} [h(t) - h(t-1)]\mathrm{d}t$$



Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!